

Chapter 2. Financial Data Ecosystem

As discussed throughout this book, data will be placed at the heart of financial market operations. As digital transformation and cloud migration efforts make their way through the financial arena, more and more financial activities will leave a digital trace. Consequently, the amount, variety, and speed of financial data generation is constantly increasing. This, in turn, has given rise to a vast and complex financial data landscape, encompassing numerous sources, types, structures, providers, delivery methods, and datasets essential for market participants to conduct their financial and business activities.

This chapter breaks down the key components of the financial data ecosystem and illustrates their features and issues.

These include the following key areas:

- The different sources of financial data
- The data structures used in finance, along with their technical details
- The variety of data types that the financial industry generates and consumes

Then, I'll give an overview of some of the most important financial datasets used by practitioners and researchers.

Sources of Financial Data

Financial data can originate from a variety of sources, so let's establish a few baselines. In this book, the term data source will be used to indicate the location, entity, or mechanism behind the generation of financial data. These sources may differ in a number of ways:

Accessibility

Whether the data is open access, commercially licensed, or proprietary

Usability

Whether the data source is easy to use and extract data from, and whether the data is provided in a raw format or in a clean structure that is ready to use

Coverage

Whether the data source offers complete and wide coverage of financial data

Reliability

Whether a source is trusted, secure, timely, and durable

Quality

Whether the data provided by the sources is of high quality (i.e., absence of errors, biases, duplicates, etc.)

Documentation

Whether additional information is available on the data structures, fields, and content of the data source

There are many different data sources in the financial data ecosystem. In this book, I categorize them into six main types: publicly available data, data generated by security exchanges, commercial data provided by vendors and distributors, survey data, alternative data, and confidential data. Let's explore the main features of each data source in detail.

Public Financial Data

Public data is information that is freely accessible by anyone, without a license or fees, primarily through the internet. Public financial data exists for a number of reasons, such as regulatory disclosure and reporting requirements, publicly released data by governmental and nongovernmental

tal organizations, public research data, and free stock market APIs. Let's explore each of these sources in detail.

Regulatory disclosure requirements

Many financial institutions are subject to data disclosure and reporting obligations established by national and international regulations. Such requirements aim to increase market transparency, efficiency, investor protection, and liquidity, as well as reduce the cost of capital for firms. For example, in the United States, companies are required to submit different reports about their operations to the Securities and Exchange Commission (SEC) as established by the Securities Act of 1933, the Securities Exchange Act of 1934, the Trust Indenture Act of 1939, and the Investment Company Act of 1940.

A variety of SEC public forms are [available](#) for companies to file periodic financial statements and various other disclosures. The most common form is the *10-K*, which is filed annually and provides a comprehensive overview of the company's financial performance. The *10-Q*, a comparable but less detailed document, is submitted following each of the three quarters for which a 10-K filing is not required. A special form, called *8-K*, is used to notify investors or the SEC about special and relevant events that might occur between 10-K and 10-Q filings. Another well-known type of disclosure is the *prospectus*, which is a document providing details about a stock, bond, or mutual fund investment offering to the public.

The main advantage of regulatory disclosures is that they constitute a reliable, unbiased, and regular source of data, covering a wide range of regulated entities. The main challenge, however, is that reporting data is often provided in raw and unstructured or semi-structured formats (e.g., HTML, TXT, or XML). Another issue is the potential for incomplete data due to the tradeoff between market efficiency and disclosure precision, which arises from regulators' demands that businesses reveal just enough data to ensure market efficiency without jeopardizing competition.¹

EDGAR is the primary data management system used by the SEC, designed to provide a seamless experience for businesses and individuals required to file reports with the agency. The EDGAR Business Office (EBO) is the official owner of EDGAR and is responsible for its administration and maintenance, strategic planning and development, and SEC staff customer support. The EDGAR database is public and free, providing access to millions of company and individual filings. The system handles roughly 3,000 filings per day, serves about 3,000 terabytes of data annually, and accommodates an average of 40,000 new filers per year.

Public institutional and governmental data

A number of national and international financial institutions regularly collect, curate, and freely publish financial and economic data. Such institutions are often regulatory or multinational agencies. Examples include the following:

World Bank Open Data

Published and maintained by the World Bank, this resource provides a large collection of financial and macroeconomic indicator datasets.

Data.europa.eu

Published by the European Union, this site provides a central point of access to open data published by European institutions and national data portals. It provides access to 1,536,561 datasets, grouped into 179 catalogs and originating from 36 countries.

The ECB Statistical Data Warehouse

Published by the European Central Bank (ECB), this resource provides data on all Eurozone economic, financial, and monetary statistics released by the ECB and Eurostat in relation to monetary policy.

Published by the Federal Reserve Bank of St. Louis, this database provides a central point of access to hundreds of thousands of economic data time series collected from national, international, public, and private sources.

Public research data

Researchers in financial markets may create and make publicly available research-oriented datasets that can be freely accessed and used for further empirical investigation. For example, Professor Kenneth French has created an online library, the [Ken French Data Library](#), that provides public and freely accessible financial data on US monthly Fama/French 3 factors and 5 factors, as well as other economic and financial variables.

Another example is datasets made public on Kaggle, where researchers and machine learning practitioners publish their models alongside the datasets that they used for training.

Free stock market APIs

A number of commercial companies, both financial and nonfinancial, offer free online financial data services such as news, prices, quotes, press releases, financial reports, and more. Examples include Yahoo Finance, Google Finance, and Alpha Vantage. Such services frequently have APIs that enable programmatic access to financial data (e.g., [Yahoo's API](#)).

WARNING

While free APIs provide convenient and cost-free access to financial data, I recommend using them with caution. For classroom-level financial analysis or experimental purposes, it's perfectly fine to use such data. However, for real financial applications and academic research, the data might lack the necessary quality guarantees. Potential quality issues may include errors, lack of adjustment for corporate actions, biases, incompleteness, and lack of proper identifiers.

Companies like Yahoo and Google do not prioritize financial data services as a key element of their competitive strategy, so they may not be motivated to offer the highest quality financial data for free.

Security Exchanges

Market exchanges like the London Stock Exchange, New York Stock Exchange, and Chicago Mercantile Exchange gather and manage diverse datasets on the transactions they handle. Being the hub of transaction activities, these exchanges provide detailed and up-to-date data crucial for trading and investment purposes. Additionally, they maintain historical archives and reference data, offering valuable insights for analysis.

It's important to note that not all market trading takes place on official exchange venues. Some financial products, like bonds and various derivatives, are traded over the counter (OTC). This term refers to trading conducted off-exchange through broker-dealer networks. Consequently, statistics on OTC transactions are often unavailable or challenging to gather and access.

Commercial Data Vendors, Providers, and Distributors

One of the most valuable and reputable sources of financial data are commercial companies specializing in financial data and software. These providers collect and curate data from a large number of sources, including regulatory filings, news agencies, surveys, company websites, brokers, banks, rating agencies, company reports, bilateral agreements with companies, exchange venues, and many more.

Morningstar, Inc., is a well-known financial data vendor that collects, analyzes, and provides financial market data, in particular stock and fund data. In one of their 8-K filings, Morningstar experts provided a detailed overview of their data collection and acquisition process. Here's an excerpt from the original filing:

We [Morningstar] collect most of our data from original source documents that are publicly available, such as regulatory filings and fund company documents. This is the main source of operations data for securities in our open-end, closed-end, exchange-traded fund, and variable annuity databases, as well as for financial statement data in our equity database. This information is available at no cost.

For performance-related information (including total returns, net asset values, dividends, and capital gains), we receive daily electronic updates from individual fund companies, transfer agents, and custodians. We don't need to pay any fees to obtain this performance data. In some markets we supplement this information with a standard market feed such as Nasdaq for daily net asset values, which we use for quality assurance and filling in any gaps in fund-specific performance data. We also receive most of the details on underlying portfolio holdings for mutual funds, closed-end funds, exchange-traded funds, and variable annuities electronically from fund companies, custodians, and transfer agents.

...Our separate account and hedge fund databases require more specialized information, which we obtain by sending surveys to the management companies. We also survey for some specialized portfolio and operations data in our other databases to enhance our proprietary portfolio statistics. We supplement information gathered electronically or through surveys by licensing a few third-party data feeds for market indices and individual securities.

As you can see, financial data vendors rely on a variety of data sources in their data collection process. This is done to supplement and enhance their data offers. Clearly, financial data vendors are continually expand-

ing their offerings and services by including new and larger datasets derived from new sources.

Compared to other data sources, commercial datasets offer the following advantages:

- Providing structured and standardized data that is highly suited for analysis, thus reducing the need for extensive data cleaning and preprocessing
- Enriching financial datasets with additional fields and identifiers for better analysis
- Providing comprehensive documentation on data usage and field metadata
- Providing a wide range of data delivery options and formats that can suit various business applications
- Providing customized solutions and packages that fit different business needs
- Providing customer support

There is a large variety of financial data vendors, some of which create their own content (often called data providers); others specialize in aggregating or distributing data, while others act as both distributors and creators of financial data. Generally speaking, financial data companies compete along the following axes:

Data coverage

The universe of financial instruments, entities, sectors, and variables is quite massive. Some vendors focus on a subset of financial data (e.g., asset classes, geographical areas, and sectors), while others tend to act as global aggregators or serve as a one-stop shop, offering a breadth of financial data coverage (such as prices, quotes, news, press releases, macroeconomic data, ratings, and volatility indicators).

Delivery mechanisms

Different providers offer their data via one or more delivery mechanisms. These may include SFTP (Simple File Transfer Protocol), cloud, data feed, API, desktop, web access, and others.

Delivery formats

Providers can differ in the file formats they use for data delivery. Examples include CSV (comma-separated values), XML, HTML, JSON, SQL query, and Zip Archive.

Data history

Some providers, especially older ones, could have longer historical coverage than others. Additionally, some providers might provide point-in-time snapshots of their data, while others provide only the latest data releases.

Delivery frequency

Data can be provided continuously (real time) or with fixed frequency such as second, minute, day, market closing time, week, or month.

Data standardizations and adjustments

Some providers might apply data transformations or standardization, while others would deliver the data in its original format. Here are a few examples:

Adjusted stock prices

A vendor might deliver adjusted stock prices that take into account corporate events such as stock splits and dividends, while another provider would leave the data unadjusted.

Data aggregation

Some vendors provide data at the exchange level, while others aggregate the data across exchanges. If you are interested in trading on a specific exchange, you might want to check if a data provider sells data for that exchange.

Standardization of accounting figures across countries

This might impact analysis if a company uses a specific accounting method that gets lost during standardization.

Data reliability and quality

Accuracy, quality, and timely access to financial data are crucial to financial institutions. Providers who guarantee high data quality, low error and bias rates, and high availability are likely to be more trusted.

Value-added services

Providers who enrich the data with extra fields, identifiers, documentation, and customer support are likely to have a competitive advantage.

Pricing

Many subscription plans are available, and they vary by vendor. Some offer large packages that can be more expensive, while others offer various package sizes that can fit multiple usage patterns and needs.

Technical limitations

Vendors might impose certain limits and quotas on their servers, such as the maximum daily requests, the maximum number of instruments that can be queried, or the request timeout.

The market for financial data is competitive and innovative, and it is continuously evolving to accommodate new products, markets, technologies, data types, and delivery mechanisms. One of the winning strategies that financial data vendors seem to invest in is the *one-stop shop* model, where a financial data vendor provides an integrated platform with access to a wide range of financial data as well as complementary services such as analytics and insights, export options, artificial intelligence tools, visualizations, messaging and chatting, trading, and search capabilities. Another competitive strategy is the *network effect*, where the value of a data vendor's solution increases as more people use it. The more users

and traders engage with a specific data vendor platform, the more appealing it becomes for new customers to join.

A significant share of the financial data market is dominated by a few players. This includes Bloomberg, LSEG, FactSet, S&P Global Market Intelligence, Morningstar, SIX Financial Information, Nasdaq Data Link, NYSE, Exchange Data International (EDI), Intrinio, and WRDS. There are also smaller players that tend to offer innovative products focused on a specific market niche. For example, firms such as PitchBook and DealRoom provide private market data on startups, private equity, and venture capital.

Next, I'll present an introductory overview of some of the most important providers in the market.

Bloomberg

Bloomberg is the whale in the financial data market, comprising almost one-third of the total market share. Bloomberg's flagship product is the [Bloomberg Terminal](#), a computer system well-known for its black interface that provides users with access to real-time market data, news, quotes, insights, and a number of valuable complementary services. Bloomberg is best suited for buy-side and asset management professionals such as traders and portfolio managers.

A valuable feature of Bloomberg is Instant Bloomberg (IB), a messaging service that allows users to chat with a large pool of financial professionals who are using the Bloomberg Terminal. Additionally, users of the Bloomberg Terminal can place trading orders via an end-to-end secure trading service (Tradebook). Recently, Bloomberg introduced BloombergGPT, an AI-powered language model that helps financial professionals with challenging language tasks such as sentiment analysis, news classification, question answering, and more. Additionally, Bloomberg provides [API services](#) that allow developers to access Bloomberg data programmatically using various programming languages.

LSEG Eikon

[LSEG Eikon](#) is Bloomberg's main competitor, and it holds a significant market share. Similar to Bloomberg, Eikon has a rich collection of financial datasets, a feature-rich user interface, developer APIs, an instant messaging service (LSEG Messenger), and trade execution capabilities. Compared to Bloomberg, Eikon has much cheaper options for more limited offerings.

FactSet

[FactSet](#) offers an affordable solution to access real-time financial data, combining proprietary, third-party, and user data. Some of FactSet's advantages include the user-friendly UI, PowerPoint integrations for PitchBooks, personalization options, a large variety of alternative datasets, and portfolio analysis tools.

S&P Global Market Intelligence

[S&P Global Market Intelligence](#) is a leading financial data and market intelligence service provider. It provides financial and industry data, analytics, news, and research. Among the most well-known solutions of S&P GMI is Capital IQ, a web-based platform that provides a rich set of data points on company financials, transactions, estimates, private company data, ownership data, and more.

Wharton Research Data Services

[Wharton Research Data Services \(WRDS\)](#) is the leading platform in financial and business research and analysis and is among the most popular data distributors in finance. WRDS offers a globally accessed, user-friendly web interface with more than 600 datasets from more than 50 data vendors across multiple domains, with a particular focus on finance. Through WRDS, users can access multiple data sources, documentation, analytics, and query-building tools simultaneously. WRDS establishes distribution and resale agreements with data vendors and allows their clients to access vendor data directly through their platform. For some

datasets, WRDS requires their clients to buy and maintain a separate license for the data.

Most financial institutions use services from at least one financial data vendor. However, finding and choosing the right vendor can be quite challenging. My advice is to start by formalizing your company's or project's financial data needs. Then, using the vendor differentiating criteria mentioned previously, you can ask yourself some key questions:

What type of data am I looking for?

For example, if you are looking for data on Chinese stock prices, you might want to find a local data provider specializing in the Chinese market.

Which fields are mandatory, and which are optional?

For example, if you want adjusted stock prices, you need to choose a data provider that provides already adjusted data or the necessary fields to perform the adjustment.

What is the universe of assets that I want?

For example, if you want data on European and American stocks, as well as fixed-income data, you might need a global data provider with international coverage (e.g., Bloomberg).

Do I need data quality guarantees?

Some providers ensure their data is free from errors, biases, and other quality issues, while others offer less assurance, necessitating data cleaning.

What is the planned budget?

For example, if your company has a limited budget, you might need to look for a data provider that offers small packages or on-demand pricing.

What purpose do I need the data for?

For example, buy-side professionals (e.g., traders) often need a wide variety of data, such as prices, news, macroeconomic vari-

ables, and market volatility, while sell-side professionals (e.g., investment bankers) need more specific data, such as asset prices and corporate events.

What data update frequency do I want?

For example, if you have a machine learning model that you train weekly, then there is no need for very frequent updates. However, if you are working at an algorithmic trading company, then data would be needed at a very high frequency.

Do I need specific delivery mechanisms?

For example, if you are developing a web application, you will very likely need to fetch data via an API. However, if you have data extraction running in the background, receiving files via SFTP or direct website download should work fine.

Do I need predictable delivery performance?

Some providers may offer guarantees on latency, throughput, and uptime, which are crucial for time-sensitive applications.

Can I tolerate vendor limitations?

For example, if your workload is predictable, you can adapt your application and consumption patterns to the limitations established by the data vendor. However, if your data needs are unpredictable or growing, you might want to find a data vendor that does not impose strict limitations or negotiate custom quotas.

Do I need a simple solution?

For example, if you want a simple user interface with Microsoft Excel or CSV export capabilities, you may not want to choose a very sophisticated financial data solution.

Survey Data

A survey is a set of questions designed to collect information from a specific group of people on a particular topic. Survey data can enhance exist-

ing datasets and provide deeper insights. For instance, banks might ask clients to complete a survey to assess their financial knowledge and risk appetite level. This information can help assess and tailor investment plans to fit each customer's profile. Similarly, financial data vendors may rely on surveys as a mechanism to gather data from companies by sending them questions related to their operations, performance, and structure.

Survey data could serve as the main ingredient in creating other types of data. For example, the Institute for Supply Management (ISM) publishes the [*Purchasing Managers' Index \(PMI\)*](#), a monthly figure used by market agents as an indicator of the direction of the economy. To calculate the PMI, the ISM sends a survey to a list of companies that make up a representative sample of the entire economy. Once respondents return the filled-in surveys, the index is computed by aggregating and weighing the individual responses.

The main advantage of survey data is its flexibility, enabling the questioner to design the questions to guarantee valuable answers and organize the information optimally. However, a few challenges might arise. For example, if the survey is voluntary, some respondents might have no incentive to provide answers, which can bias the final outcome. In other cases, some candidates may choose not to complete the survey. This could be due to a reluctance to report poor financial performance, organizational inertia, concerns about reputational risk, competition, or issues related to security and confidentiality. Additionally, the questioner might (intentionally or unintentionally) induce *framing bias* in the survey by choosing questions intended to force the respondent to give a desired answer. For this reason, a strong emphasis on ethical practices and bias checks is highly recommended when conducting surveys.

Alternative Data

In recent years, the term “alternative data” has emerged in the financial world to describe data sources not traditionally used in financial analysis and investment. These nonconventional data sources are not inherently embedded within the financial system, unlike trading venues, payment

systems, and commercial banking. The sources of alternative data are quite heterogeneous. A few are mentioned here:

- Satellites (e.g., collecting shipping images)
- Social media (e.g., tweets)
- Articles
- Blog posts (e.g., Medium, Substack)
- News channels (e.g., Thomson Reuters, Bloomberg)
- Weather observers
- Online clicking patterns and browsing history
- Shipping and pipeline trackers
- Emails and chats
- Product and service reviews
- Security, social, and environmental scores and ratings

Confidential and Proprietary Data

Financial institutions generate and store large amounts of data when conducting their internal operations. The most essential and valuable type of internal data involves contextual, nontransactional information about the various business entities. This includes, for example, data about clients, employees, suppliers, products, subscriptions, offerings, discounts, pricing, financial structures (e.g., cost centers), and locational information (e.g., branches).

A second type of internal business data is *transactions*. In performing their daily activities, financial institution clients generate a large amount of transaction data, such as payments, deposits, trading, purchases, credit card transactions, and money transfers.

Finally, financial institutions might collect *internal analytical data* from research teams and analysts conducting market and financial analysis. This includes, for example, sales analysis and forecasting, customer segmentation, credit scoring, default probabilities, investment strategies, stock predictions, macroeconomic analysis, and customer churn probabilities.

In specific situations, financial institutions must report various types of confidential data to regulatory bodies. This includes trade details, large transaction information, suspicious activity reports, data gathered during anti-money laundering (AML) and know your customer (KYC) processes, as well as information related to tax, compliance, liquidity, risk management, and capital adequacy.

Now that you have an idea about the different sources of financial data, the next section will discuss the various structures in which financial data can be represented.

Structures of Financial Data

Financial data can be stored and represented using various structures. These may range from simple data structures such as tabular times series, cross-section, or panel data to more complex structures such as matrices, graphs, and text. Throughout this section, we will explore these structures in detail.

Time Series Data

A time series is a collection of records for one or more variables associated with a single entity, observed at specific intervals over time. Time series are indexed in time order and can be represented mathematically as follows:

$$\mathbf{X} = \{X_1, X_2, \dots, X_N\}$$

where N is the length of the time series and X is the observed variable, or:

$$\mathbf{X} = \{X_t, t \in T\}$$

where T is a time index set and X is the observed variable.

In tabular format, a time series with S variables observed over N periods can be represented as shown in [Table 2-1](#). The first column stores the temporal dimension (T_N), while the other columns (X_S) store the time

series values for each variable. A single cell (X_{SN}) records the time series values observed at a specific time for a given variable.

Table 2-1. Tabular representation of time series

Time/variable	X_1	X_2	X_3	...	X_S
T_1	X_{11}	X_{21}	X_{31}	...	X_{S1}
T_2	X_{12}	X_{22}	X_{32}	...	X_{S2}
...	...	...	...	...	...
T_N	X_{1N}	X_{2N}	X_{3N}	...	X_{SN}

The time series is the most common data structure in financial markets, mainly due to the dynamic and transactional nature of financial activities that happen over time. Examples of financial activities that generate time-series data include trading, pricing, payment, investment, estimation, valuation, risk measures, volatility figures, corporate events, performance and accounting, and many more.

Extensive literature on financial time series has been produced where a large number of temporal phenomena have been investigated. For an excellent treatment of financial time series analysis, I recommend the seminal book *Analysis of Financial Time Series* by Ruey Tsay (Wiley, 2010).

Cross-Sectional Data

A cross-section is a collection of records for one or more variables observed across multiple entities at a single point in time. In cross-sectional data, the time series dimension is irrelevant, and the emphasis is on the variables themselves. Entities in a cross-section often share common characteristics, such as firms within the same sector, investments in a specific strategy, fund managers, and more.

A cross-section dataset can be represented mathematically as follows:

$$X_{t=z,i,j}, i = 1, \dots, N, j = 1, \dots, S\}$$

where t is the time index (fixed), i is the entity index, and j is the variable index.

In tabular format, we can draw a cross-section of N entities and S variables as shown in [Table 2-2](#). The first column stores the names of the entities (E_N), while the other columns store the cross-section value of the S variables (V_S). A table cell ($X_{i,j}$) stores the cross-section value of a given entity for a given variable.

Table 2-2. Tabular representation of cross-section data

Entity/variable	V_1	V_2	V_3	...	V_S
E_1	X_{11}	X_{21}	X_{31}	...	X_{S1}
E_2	X_{12}	X_{22}	X_{32}	...	X_{S2}
...	...	...	...	...	...
E_N	X_{1N}	X_{2N}	X_{3N}	...	X_{SN}

Financial cross-section data is generated from the presence of many participants in financial markets, including financial institutions, brokers, investors, consumers, traders, and managers, as well as a variety of other entities such as financial assets, investment strategies, and consumer and firm choices and behaviors, among others.

Financial cross-sectional data can be used to understand and explain significant point-in-time differences between financial entities, such as why different stocks have different returns, why firms behave differently, or why performance across fund managers varies. Furthermore, cross-sectional analysis is a key tool for identifying correlations, causations, or anomalies across different entities, which can be critical for investment decisions, policy formulation, and understanding market dynamics.

Panel Data

Panel data combines both time series and cross-sectional structures, representing data for a set of variables across multiple entities at different points in time. Mathematically, a panel has the following form:

$X_{i,j,t}, i = 1, \dots, N, j = 1, \dots, S, t = 1, \dots, T\}$

where i is the entity index, j is the variable index, and t is the time index.

In tabular format, we can draw a panel of two entities, two time periods, and three variables as shown in [Table 2-3](#). The first column stores the names of the entities, the second column stores the time, and the last three columns store the observed panel values for all three variables. This panel representation is called *wide format*, as it expands horizontally when new variables are added to the panel. Another option is the long format, where the variable name and value are stored in separate columns, expanding the table vertically with new entries. An example is shown in [Table 2-4](#).

Table 2-3. Wide tabular representation of panel data

Entity	Time	V_1	V_2	V_3
E_1	t=1	X_{111}	X_{112}	X_{113}
E_1	t=2	X_{121}	X_{122}	X_{123}
E_2	t=1	X_{211}	X_{212}	X_{213}
E_2	t=2	X_{221}	X_{222}	X_{223}

Table 2-4. Long tabular representation of panel data

Entity	Time	Variable name	Variable value
E_1	t=1	V_1	X_{111}
E_1	t=2	V_2	X_{122}
E_2	t=1	V_1	X_{211}
E_2	t=2	V_2	X_{222}

Financial panel data arises from numerous market entities engaging in various activities over time. For instance, stocks are continuously traded and priced, companies regularly submit quarterly and annual reports, and individuals conduct daily payments and transactions. Hence, most financial datasets are essentially panel datasets.

Panel data may vary in terms of their cross-section and time series components. For example, annual balance sheets tend to have a large cross-section component (many firms) and a small history. On the other hand, some stock price panel datasets tend to have a long history (e.g., high-frequency trading prices) and a smaller cross-section component (asset universe).

An important characteristic of panel datasets is their degree of [balance](#). Assume a panel has N entities, T time periods, and S variables. The panel is said to be *balanced* if all $N \times T \times S$ observations are available; otherwise, the panel is called *unbalanced*. [Table 2-5](#) shows a balanced panel where data are available for all entities, time periods, and variables. [Table 2-6](#) instead illustrates an unbalanced panel as it has six cells with null values, resulting in half of the observations being missing.

Table 2-5. Balanced panel

Entity	Time	V_1	V_2	V_3
E_1	t=1	X_{111}	X_{112}	X_{113}
E_1	t=2	X_{121}	X_{122}	X_{123}
E_2	t=1	X_{211}	X_{212}	X_{213}
E_2	t=2	X_{221}	X_{222}	X_{223}

Table 2-6. Unbalanced panel

Entity	Time	V_1	V_2	V_3
E_1	t=1	Null	X_{112}	Null
E_1	t=2	X_{121}	X_{122}	X_{123}
E_2	t=1	X_{211}	Null	X_{213}
E_2	t=2	Null	Null	Null

Similar to time-series data, a large literature, both theoretical and empirical, has been produced for panel data analysis. For an excellent introduction, I highly recommend Badi H. Baltagi's seminal book *Econometric Analysis of Panel Data* (Springer, 2021).

Matrix Data

A matrix is a two-dimensional array of elements arranged as rows and columns. Here is an example:

$$\begin{pmatrix} 2 & 5 & 7 \\ 3 & 4 & 6 \\ 7 & 7 & 3 \end{pmatrix}$$

This is a matrix with three rows and three columns, often referred to as a 3×3 matrix.

The use of matrix structures is quite common in finance. The best example is perhaps portfolio optimization theory, in particular Markowitz's mean-variance optimization (MVO). According to MVO, when constructing a financial investment portfolio, three elements should be taken into consideration:

- The expected return on the assets in the portfolio
- The variance (risk) of the asset returns
- The covariances between the asset returns

For example, let's imagine a portfolio with three assets: A, B, and C. Let's denote the expected return of any asset i with $E(R_i)$, the variance with $var(R_i)$, and the covariance between two assets i and j with $cov(R_i, R_j)$. Using MVO, two matrix representations would be constructed as follows:

- 3×1 portfolio return matrix

$$\begin{pmatrix} E(R_A) \\ E(R_B) \\ E(R_C) \end{pmatrix}$$

- 3×3 covariance matrix

$$\begin{pmatrix} var(R_A) & cov(R_A, R_B) & cov(R_A, R_C) \\ cov(R_B, R_A) & var(R_B) & cov(R_B, R_C) \\ cov(R_C, R_A) & cov(R_C, R_B) & var(R_C) \end{pmatrix}$$

Using these matrix representations, MVO relies on matrix algebra to optimize the portfolio's expected return for a given level of risk appetite.

Portfolio optimization means choosing the best asset allocation strategy (how much to invest in each asset) to achieve a desired investment goal.²

Graph Data

Financial markets are an outstanding example of a complex system of myriad relationships, transactions, dependencies, and flows.

Understanding such complex interactions can provide valuable information and insights into market structures, systematic risks, contagion mechanisms, dominant market positions, fraudulent behavior, and market inefficiencies.

The kind of analysis that focuses on studying the complex interactions in a system is called *network analysis* or *network science*.³ To employ network science in finance, traditional time-series or cross-section data would lack the depth and granularity required to build and analyze financial networks. To this end, graph data is required. This type of data is also frequently referred to as connections, networks, or nodes and links. A graph dataset consists of two sets of data: a set of nodes (aka vertices) together with node attributes (e.g., name, country, type, and so on), and a set of links (aka edges) with link attributes (e.g., type, value, sign, and so on).

When working with graph data, an important challenge concerns the decision about how to structure the data for quick access and analysis. Structuring nodes' data and their attributes is straightforward, and a tabular structure will do the job. However, things get more complicated when it comes to structuring link data. To this end, special graph structures are often used. Generally speaking, a graph dataset can be represented in four main ways:

Network visualization

A 2-D drawing of the nodes (often as circles) and links (using straight lines). This method is useful for illustrative purposes and works best with small networks.

Adjacency matrix

An $N \times N$ matrix (N being the number of nodes). If there is a link between nodes i and j , it stores 1 at positions (i, j) and (j, i) .

Adjacency list

An array of length N (N being the number of nodes) where each item in the array contains the index of the node and a list of its neighbors represented via a linked list.

Edge list

A simple array that stores all edges of a graph. An item of an edge list is a tuple where the first element is the source node and the second is the target, with optional elements that may represent link attributes such as weight, sign, and time.

Figure 2-1 visually illustrates these four types of graph data representations.

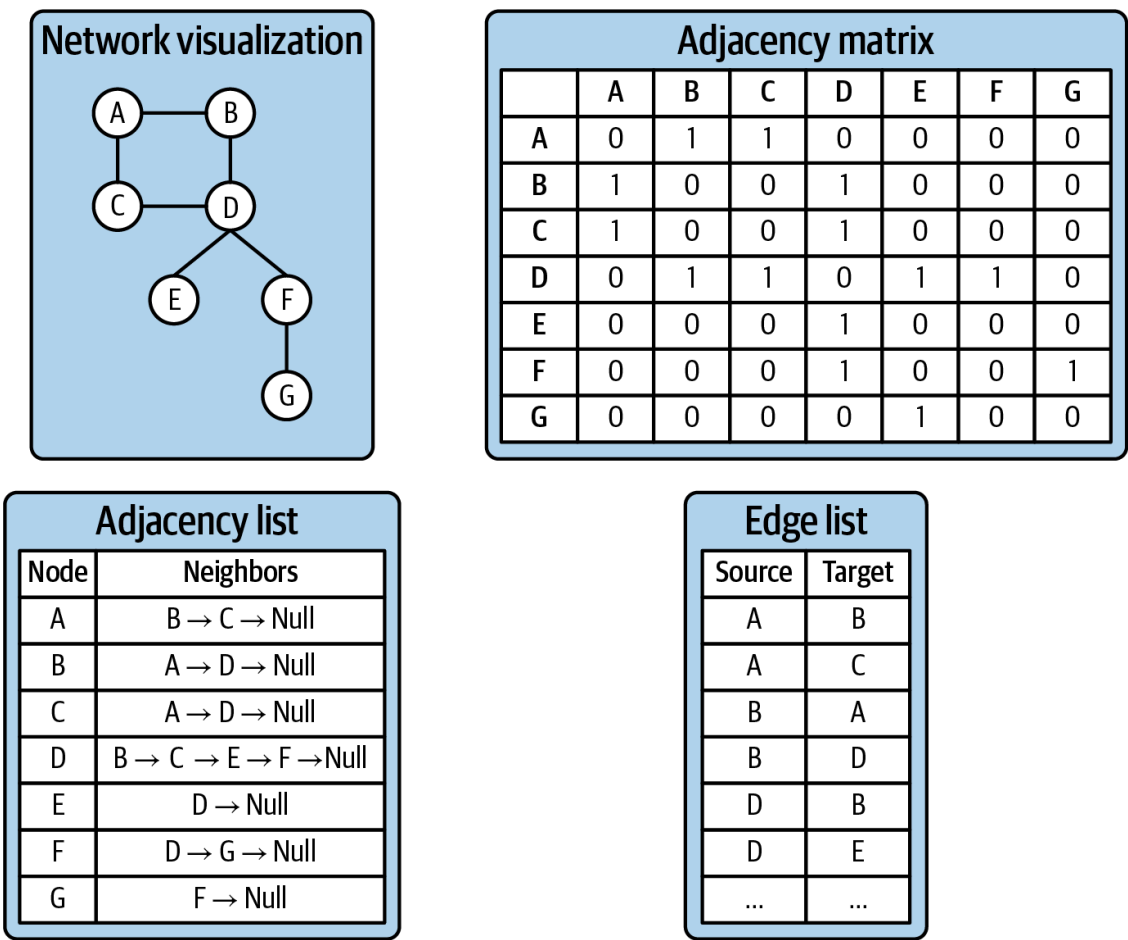


Figure 2-1. Different representations of graph data

Financial markets rely on a range of graph data types and representations. In the upcoming sections, I will discuss six pertinent graph structures: simple, directed, weighted, temporal, multipartite, and multiplex.

Simple graphs

A simple graph consists of a set of homogeneous nodes (nodes of the same type, e.g., companies) and a set of homogeneous, unweighted, and undirected links. [Figure 2-2](#) illustrates an example with two nodes (A, B).

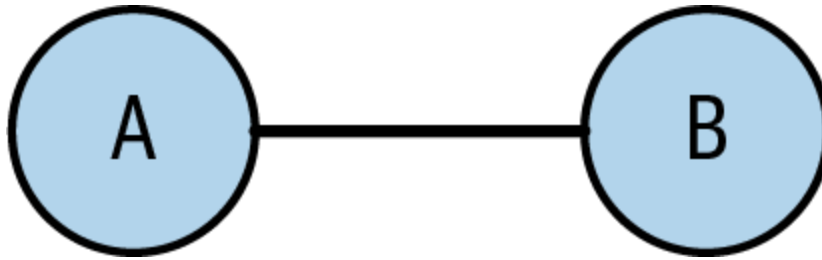


Figure 2-2. A simple graph with two nodes

The adjacency matrix, adjacency list, and edge list representations of simple graphs are similar to those shown in [Figure 2-1](#). In financial markets, simple graphs can represent the *presence* of relationships between entities, such as payment agreements between banks, partnership agreements, and statistical correlations between financial assets.

Directed graphs

In a directed graph, links have a direction indicating an *orientation* between the nodes. Directed graphs are used to represent relationships where nodes at each end of the link play different roles. For example, A borrows from B, where A is the borrower and B is the lender. [Figure 2-3](#) illustrates an example with two nodes, A and B, where node A points to node B but not vice versa.

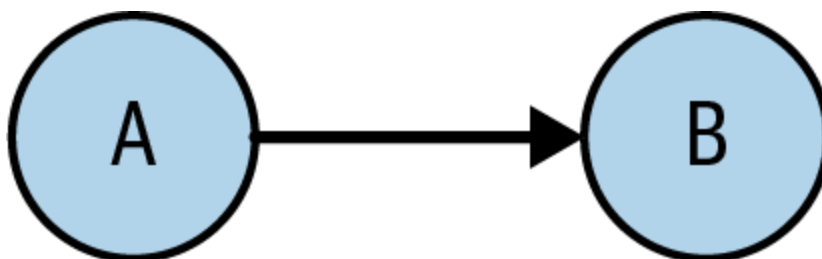


Figure 2-3. A simple directed graph with two nodes

The adjacency matrix of a directed network will have a value of 1 at position (i, j) if there is a link pointing from node i to j, but will have a value of 0 in position (j, i) if j doesn't have a link pointing toward i. Similarly, a directed graph's adjacency and edge list will only store records for nodes that point toward other nodes. [Figure 2-4](#) illustrates the four representations for a directed graph.

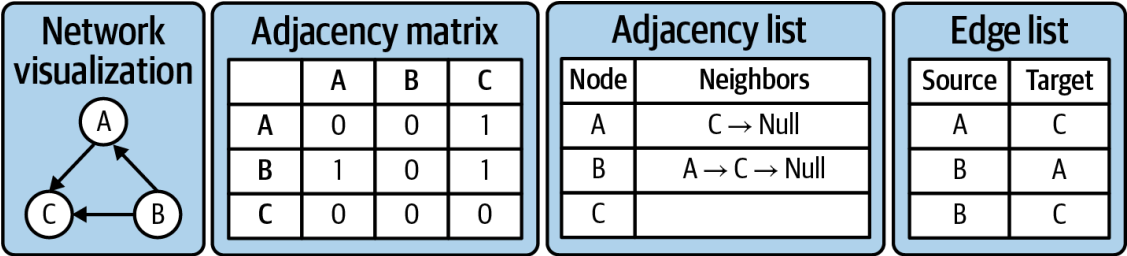


Figure 2-4. Directed graph representations

Directed graphs can represent a variety of financial activities, such as cross-holding among banks, transfers, transactions, and interbank lending.

Weighted graphs

In weighted graphs, links are assigned a numerical value to indicate the relationship's magnitude or strength. When representing a weighted graph, the weight needs to be added to the adjacency matrix, adjacency list, and edge list as illustrated in [Figure 2-5](#).

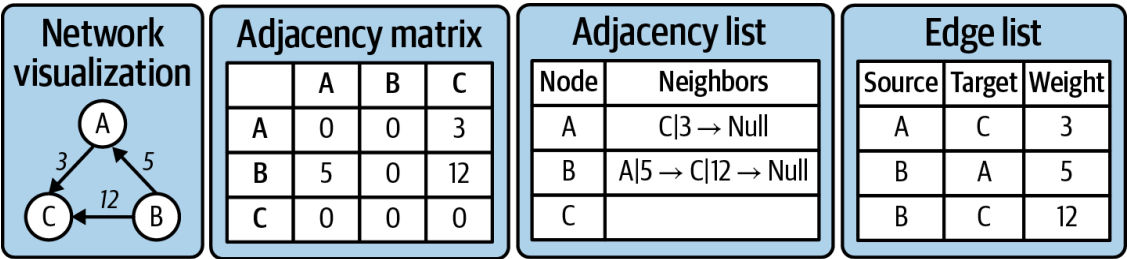


Figure 2-5. Weighted graph representations

Most types of graphs can be weighted (including simple and directed graphs). In finance, weights can represent the value of the assets one bank holds at another in a cross-holding network, the amount of money transferred in a transaction, or the number of securities sold in a market trade.

Multipartite graphs

A graph representation is called *multipartite* when it includes more than one type of node and only allows links between different types of nodes. Graphs with two types of nodes are referred to as *bipartite*, those with three types of nodes as *tripartite*, and the generalized case with k types is called *k-partite*. A *k-partite projection*, often applied to bipartite graphs, is a common operation that builds a graph that contains one type of the k available node types, and edges exist based on whether two nodes share a common link to another type of node in the original network. [Figure 2-6](#) presents a bipartite graph along with its projected graph and adjacency matrix.

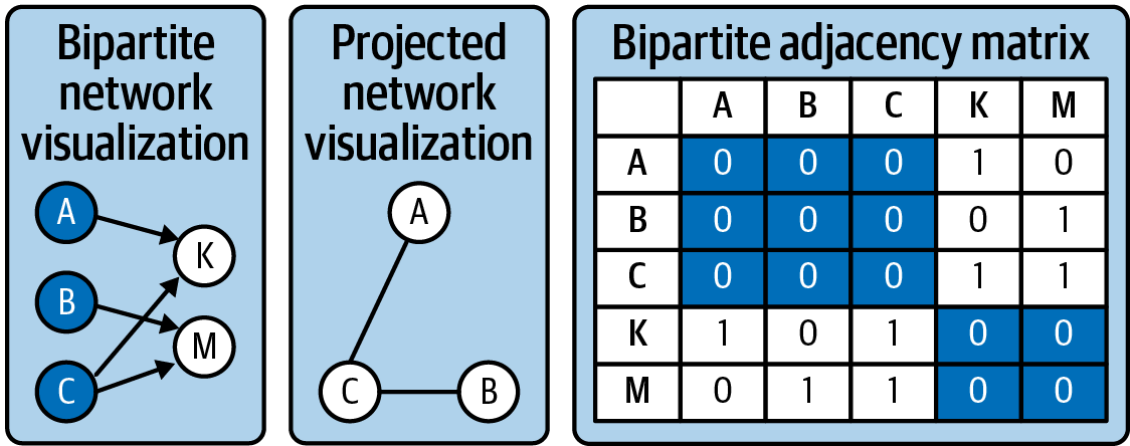


Figure 2-6. Bipartite graph representations

The leftmost graph in [Figure 2-6](#) has two types of nodes, shaded and white, and links exist only between a shaded and a white node. The middle graph illustrates a bipartite projection on one type of node (the shaded nodes in the leftmost graph). The projected graph shows that nodes A and C have a link because in the bipartite graph on the left, both A and C share a connection to K. The same goes for C and B, which share a connection to M. The adjacency matrix in [Figure 2-6](#) has a block structure, where links between the same type of nodes will be absent and represented with zeros (shaded blocks) whereas links between different types of nodes exist in the white blocks.

Examples of bipartite relationships in finance include the following:

Interlocking directorate

Nodes of type “person” (node type 1) act as a member of the board of directors (link) of one or more firms (node type 2).

Syndicated lending

Multiple lenders (node type 1) jointly provide a loan (link) to one or more borrowing entities (node type 2).

Corporate control hierarchies

Parent firms (node type 1) have ownership rights (link) over child firms (node type 2).

Correspondence banking

A correspondent banking relationship is an arrangement between two banks, typically in different countries, where one bank (the correspondent bank) provides services on behalf of another bank (the respondent bank).

Working with multipartite graphs is much easier when nodes are labeled by category, often referred to as a colored graph in mathematics. For instance, a bipartite graph can be colored with two colors, where each node category has the same color. Without such labels, identifying a k-partite graph can be challenging, especially when k is greater than two.⁴

Temporal graphs

In a temporal graph, links have a temporal dimension indicating the time at which they were active. The network representation of temporal graphs is snapshot based or time based. As illustrated in [Figure 2-7](#), both the network and adjacency representations are temporal snapshots storing the state of links at each time period. For example, nodes A and B established a link at times t1, t3, and t4, but they didn’t interact at time t2. The adjacency matrix is represented as a multidimensional matrix storing a snapshot for each time period. The edge list stores all links and the time at which they were active.

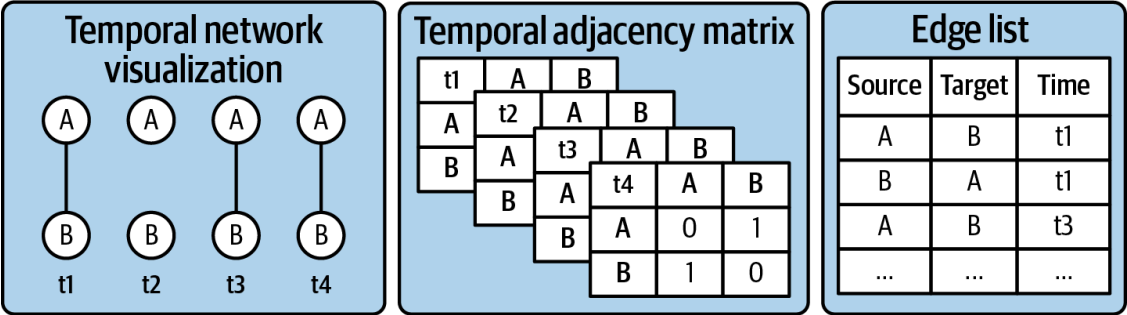


Figure 2-7. Temporal graph representations

Many financial relationships are temporal in nature, such as interbank lending, trading, payments, transactions, and many more.

Multilayer graphs

A multilayer graph is a complex structure used to represent relationships between different types of nodes and links. The most common type is the *multiplex* graph, where nodes are of the same type but have different types of links depending on the interaction context. For instance, a financial institution may operate in various markets such as commercial banking, wealth management, and financial consulting.

Multilayer network representations are very useful in finance. They can reveal potential cascading mechanisms that might amplify a local shock, help understand power structures spanning multiple areas, and enhance fraud detection (e.g., when the same individual operates in different systems using the same connection patterns). An illustration of multiplex graph representation is provided in [Figure 2-8](#).

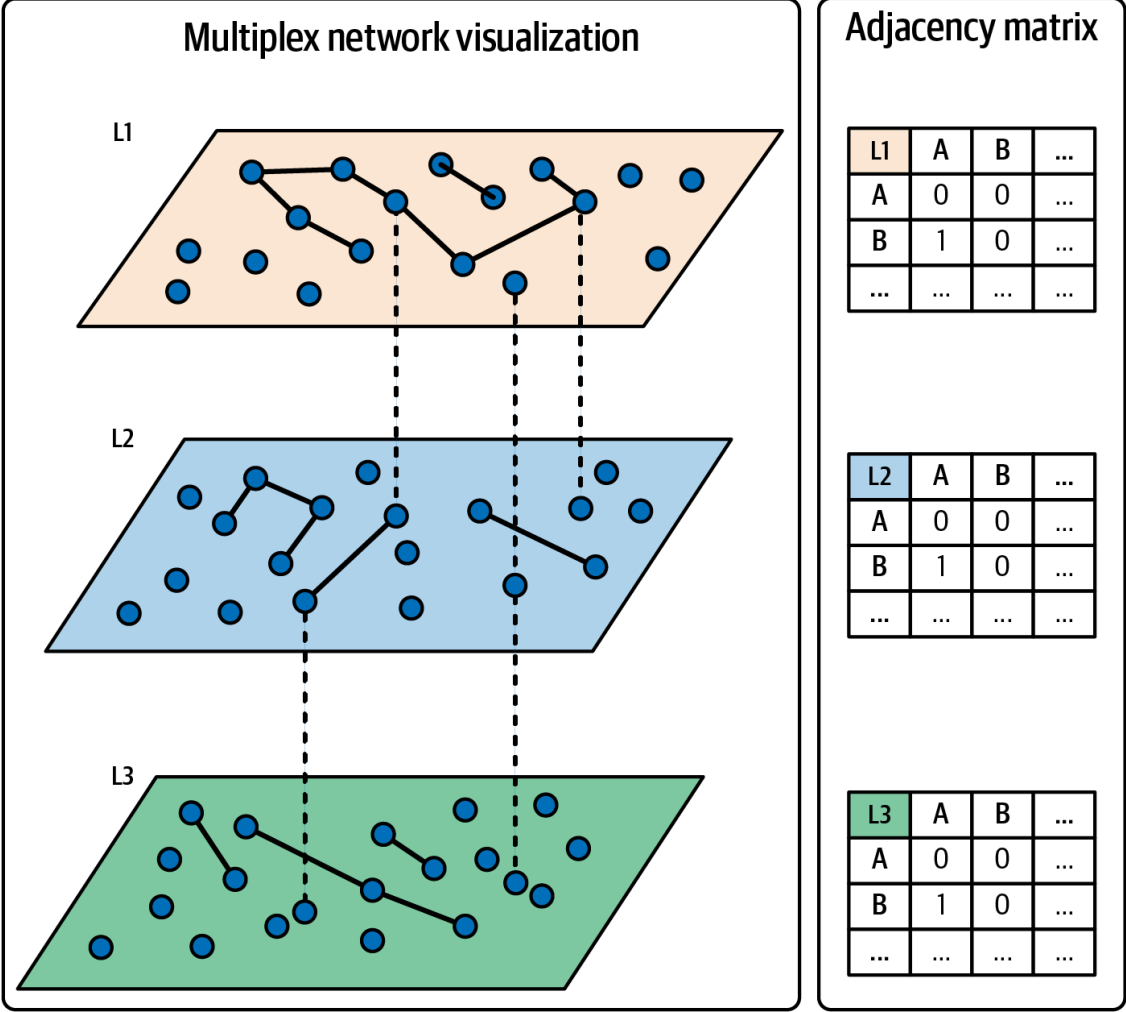


Figure 2-8. Multiplex graph representation

WHAT ARE THE SOURCES OF FINANCIAL GRAPH DATA?

An important feature of financial graph data is that it is often derived from other types of data sources. For example, to build a graph of the [interbank loan market](#), you would need a historical panel of interbank transactions containing information on borrowers, lenders, and the transaction amount and time. Similarly, say you want to build a bipartite network of [interlocking directorates](#) to see which directors sit on multiple boards, then you would need a dataset (e.g., the LSEG Officers and Directors dataset) that contains information on executive individuals and the companies they are associated with. In some cases, more work is required. For example, if you want to build a network of [similarities between stocks](#), you would first need a historical panel of price data (e.g., CRSP), and then you need to compute the pair-wise similarity (correlation) between the stocks in order to get the links needed to build the graph.

It is possible in some cases to find connection datasets curated by financial data vendors. For example, S&P Global [provides a number of datasets about business and company relationships](#), such as ownership, supply chain, investment, and many more types of relationships.

Text Data

Text data refers to information available in any unstructured or semi-structured format. Its extensive use and versatility make it [arguably the most available type of data in financial markets](#). With the advent of generative AI and large language models (LLMs), the importance of text data has significantly increased. Generative models such as GPT-4 and BloombergGPT are trained on massive amounts of text data. For specialized financial tasks such as fraud detection, sentiment analysis, know your customer (KYC), and anti-money laundering (AML), finance domain-specific text data is essential for customizing and fine-tuning an LLM.

It can be any text-based medium, including news, analyst reports, company filings, prospectus documents, emails, social media posts, system

logs, patent documents, legal and technical documents, research papers, web pages, blog posts, and much more.⁵

Text data can take various forms, including plain text consisting of words and sentences, semi-structured documents with keys and values, tabular formats, or complex specialized data structures like [word embeddings](#).

Financial text data exhibits certain characteristics that set it apart from other forms of textual data. Terms such as “liability,” “risk,” “default,” and “exposure” may carry negative connotations in certain contexts, yet they are commonly used and frequent terminology in financial markets. Additionally, financial documents can contain both explicit content (such as a firm merging with another) and implicit content (factual information implying a positive or negative emotion).⁶

Now that you’re familiar with financial data’s main sources and structures, the following section will explore the various types of financial data employed for analytical and operational purposes.

Types of Financial Data

The financial data structures we discussed in the previous section can represent a variety of financial variables, content, and phenomena. This section will categorize these financial phenomena by discussing the most crucial types of financial data in detail, illustrating their features and challenges.

Fundamental Data

Financial fundamental data, also called financial statements data, balance sheet data, or corporate finance data, conveys information about firms’ structures, operations, and performance. Firms themselves generate and produce fundamental data, partly because it’s a regulatory requirement and partly because it helps the management understand the financial and operational situation of the firm. Many companies have accounting departments liable for generating and maintaining financial statements, but it is also possible to do so via accounting service

providers such as consulting and audit firms. In general, there are three main financial statements:

Balance sheet

Provides figures on what a firm owns (assets), what it owes (liabilities), and what its shareholders own (*shareholders' equity*)

Income statement

Provides figures on a firm's financial performance over a specific period, such as the annual revenues and net profit

Cash flow statement

Provides information on a firm's cash movements (in and out), which can help determine whether the firm is generating enough cash to carry out its operations

Each financial statement contains a large number of items representing different quantities. A special type of financial statement item is *ratios*, which represent relationships between two or more items. For example, a popular ratio is *Return on Equity* (ROE), which is calculated by dividing the company's net income by the value of its shareholders' equity. A high ROE is an indicator that the company is efficient at generating net income using shareholders' equity.

Financial fundamental data statements have a number of characteristics that need to be taken into account. First, due to the time it takes to prepare them, fundamental data statements are released with *low frequency*, such as quarterly, each semester, or yearly. Second, fundamental data reports are often published with a *time lapse*, meaning that data for a specific period is released at a future date. Data can be reported in the form of *reinstatements*, which happens when a figure gets revised and corrected after it has been released. Data *backfilling* can also take place, which happens when a firm and its entire fundamental data history get added to a dataset that has never had information on that firm before.

If not handled properly, these characteristics could lead to *non-point-in-time (PIT) fundamental data*. PIT data is data that is recorded with a time-

stamp reflecting the filing or release data. With PIT data, one can ask the questions “When was the data released?” and “What data was known at that time?” On the other hand, non-PIT data is stamped with the fiscal year-end date or the latest update date. Non-PIT data reflects the latest data release, and it gets overwritten when an update or reinstatement happens. [Table 2-7](#) illustrates the difference. With three versions of the data available, a PIT dataset would keep track of all historical snapshots (101, 110, 120), while a non-PIT dataset would show only the latest version (120).

Table 2-7. PIT versus non-PIT data

Data type	Preliminary result	Fiscal year-end release	Correction
Version	101	110	120
PIT	101	110	120
Non-PIT	120	120	120

Non-PIT data poses several challenges. First is the problem of *look-ahead bias*, which happens when conducting historical studies and assuming that data was known at a specific historical moment, which, due to a release time lapse, may not be the case. Second is *reproducibility*, which becomes challenging if non-PIT data used for past research is updated and may not yield the same results. The third challenge is *transparency*, which may be an issue if, for example, a company updates an accounting record to hide past fraudulent behavior.

Being able to understand the main items in an annual report is a good skill, even for financial data engineers. The table below illustrates the annual financial highlights of an imaginary bank, ADK. If you check the annual report of any real bank, you will find a similar table. Examining the figures for 2020, we can see that the bank realized more *total net revenues* compared to 2021 (\$bln 102 versus 99) but with a lower *net income* (\$bln 23 versus 25). This translates into lower returns for shareholders, which we can see from the *return on common equity* figure of 11% compared to 16% in 2019. ADK is quite a large bank, which we can see from the *total assets* figure that equals \$2,987,245 million (almost \$3 trillion) and the *headcount* of 150,432. Moreover, ADK seems to be financially stable, which we can infer from the *Tier 1 capital ratio* of 15%, a measure introduced by the Basel III accord, which tells how stable a financial institution is by comparing its core equity against its risk-weighted assets. Basel III established that the Tier 1 capital ratio must be at least 6%. ADK probably has a large market share, which we can see from its considerable market capitalization of \$240,876 million (roughly \$240 billion). A dividend was distributed every year to shareholders, which we know from the *cash dividends per share* figure.

Item	2020	2019	2018
Total net revenue (\$ mln)	102,000	99,340	96,345
Total noninterest expense (\$ mln)	55,000	49,000	45,865
Net income (\$ mln)	23,100	25,987	19,628
Cash dividends per share (\$)	3.5	2	1.7
Return on common equity	11%	16%	8%
Tier 1 capital ratio	15.0%	14.1%	14%
Total capital ratio	18.9%	17%	17%
Loans (\$ mln)	986,865	956,586	916,865
Deposits (\$ mln)	1,298,456	1,187,456	1,062,456
Total assets (\$ mln)	2,987,245	2,445,853	2,200,119
Total stockholders' equity (\$ mln)	72,876	71,664	70,345
Market capitalization (\$ mln)	250,876	300,897	280,986
Headcount	150,432	144,983	139,709

Market Data

Market data includes price and related information for financial instruments traded on market venues such as the New York Stock Exchange (NYSE) or in off-exchange OTC markets. It is extensively available for many types of instruments, such as stocks, derivatives, indexes, currencies, bonds, and funds. Moreover, market data is released frequently, often becoming available just seconds after its generation.

A financial instrument's market data may include fields such as the following:

- Identifiers such as tickers, e.g., IBM
- Trading venue code such as the exchange code, e.g., NYSE
- Last day (adjusted) closing price, e.g., \$40.30

- Current day open price, e.g., \$40.15
- Current day highest price, e.g., \$42.30
- Current day lowest price, e.g., \$39.42
- Current day price range, e.g., \$39–42.30
- Current day volume, e.g., 50,000
- Latest bid/ask prices and quantities, e.g., bid of \$40.02 × 2200 and ask of \$40.03 × 3000
- Last trade/quote date, e.g., today at 14:12:10 p.m.

NOTE

In financial markets, the term *bid* is used to indicate the price at which buyers are willing to buy a financial asset, while the *ask* is the price at which sellers are willing to sell an asset.

Some stock exchanges and media platforms enrich market data with fundamentals such as market capitalization, revenues, and net profit, as well as various financial ratios and corporate event dates, such as the latest dividend date.⁷

At the core of market data generation lies the *order book*, an electronic ledger of buy and sell orders for a specific security at a given trading venue. Order books have two representations: *Market by Order* (MBO) and *Market by Price* (MBP). MBO shows each individual order separately, detailing the price, quantity, and time of entry for each buy and sell order. In contrast, MBP aggregates orders by price level, consolidating them into price buckets to display the total quantity of orders at each price level instead of showing each order individually.

Orders in an order book can be matched through an *Order Matching System* (OMS), which uses algorithms to ensure the best execution. The two most common match algorithms are price/time priority (also known as First In, First Out, or FIFO) and pro rata.⁸ Price/time priority matches orders based on price first, and then by the time of entry at the same price level; for example, if two buy orders are placed at the same price, the earlier one gets matched first. Pro rata matching, on the other hand, matches orders at the same price level proportionally based on their

sizes; for instance, if there are two buy orders at the same price, one for 100 shares and another for 200 shares, and a sell order for 150 shares arrives, the matching will be split proportionally, giving 50 shares to the first order and 100 shares to the second.⁹

An example of an order book is illustrated in [Table 2-8](#). The example shows that the highest bid price level is \$55.11, while the lowest ask price level is \$55.13. The highest bid and lowest ask are called the *top of the book*, while the difference between them is called the *bid/ask spread* (= \$0.02). The *order book depth* is a measure of the number of distinct price levels available in an order book. Another term, the *market depth*, is often used as an indicator of liquidity and is defined as the size of market orders that can be executed without causing a large impact on the price level.

Table 2-8. Example of an order book

Buy-side		Sell-side	
Bid volume	Bid price	Ask price	Ask volume
220	\$ 55.11	\$ 55.13	1000
1000	\$ 55.11	\$ 55.14	500
50	\$ 55.9	\$ 55.66	200
20	\$ 55.55	\$ 55.66	10
560	\$ 55.1	\$ 66.68	50
...	...	...	...

Different types of orders can be submitted to an order book. For example, a *market order* is used to buy or sell at the best bid/ask price available. A *limit order* is an order to sell at a specified minimum ask price or buy at a maximum bid price. A *stop-loss order* is triggered and submitted to buy or sell if the price level reaches a predefined maximum or minimum value. A *trailing stop order* is triggered if the market price moves up or down by a specific percentage or dollar amount.

Market data exhibits several features that are worth considering. First, market data, such as prices, are computed quantities. When an investment firm submits an order to buy or sell a financial asset, the order price is often calculated using a specific valuation method. For example, company stock prices can be calculated using methods such as the net present value, earning multiples, the dividend discount model, or the discounted cash flow model.¹⁰

Importantly, when different market participants submit orders with varying prices across different markets, determining the final market price for investors becomes crucial. Ideally, the bid price is the highest among all bid prices, and the ask price is the lowest among ask prices. To enforce this rule, regulators such as the SEC passed the [Regulation National Market System \(NMS\) regulation](#), which introduced the *National Best Bid and Offer* (NBBO). With the NBBO in action, brokers are required to ensure that their client orders are executed against the best bid and ask prices, which need to be at least equal to the NBBO.¹¹

Nevertheless, not all financial markets ensure an NBBO. This is particularly true in OTC markets like foreign exchange (Forex) markets where currencies are traded. There is no enforced exchange rate for a currency pair in the Forex market. The exchange rate you will get is specific to the Forex quote provider, which can be your bank or, most commonly, a *Forex broker*. This means that the quote you see on Google for EUR/USD is not the market price, and you should not expect it to match the one you get offered when you convert money from one currency to another.

Price discreteness is another important attribute of market data. It means that the price of a listed stock cannot change by less than a specific amount known as *tick size*. For example, according to [SEC Rule 612 on the minimum pricing increment](#):

The Rule prohibits market participants from displaying, ranking, or accepting quotations, orders, or indications of interest in any NMS stock priced in an increment smaller than \$0.01 if the quotation, order, or indication of interest is priced equal to or greater than \$1.00 per share. If the quotation, order, or indication of interest is priced less than \$1.00 per share, the minimum pricing increment is \$0.0001.

Finally, a quite important feature of market data, and in particular price data, is its *nonsynchronous history*. This means that given two securities, A and B, security A could trade at least once every minute, while security B trades only once every five hours. Such discrepancies in market behavior are often due to market liquidity varying from one security to another.

Transaction Data

A financial transaction is a legal agreement between two or more parties to exchange financial securities, services, goods, risks, or commodities in return for money. Financial markets generate massive volumes of transaction data triggered by activities such as investments, payments, trading, hedging, speculation, and lending.

Most financial transactions are conducted either fully or partially electronically. Generally speaking, a financial transaction can be characterized by at least five elements: transaction specifications, transaction parties, initiation date, settlement date, and settlement method.¹² Next, we discuss each of these five elements in detail.

Transaction specifications

Before initiating a transaction, parties agree on its details and specifications. In securities exchange, this phase is often called pre-trade. To provide examples, a stock purchase transaction would specify the name and identifiers of the shares being exchanged, the quantity of each share, the price, and the currency. For financial instruments such as options on stocks, additional details would be needed, such as option type (call versus put), expiry date, and exercise price. For cross-border payments, par-

ties may need to agree on the amount, currency pair, Forex conversion rate, and any spread margin.

Initiation date

The initiation date of a transaction is the date at which parties enter into an agreement to execute a transaction.

Settlement date

The settlement date is when a transaction is finalized, and the exchange of money and transfer of ownership takes place. In securities exchange, this phase is called post-trade. Importantly, in many financial transactions, the settlement date falls on a future date, following the initiation date. Such a delay can be due to various reasons such as the transaction verification process, technological constraints, errors due to missing information, bureaucratic steps such as the issuance of a certificate of ownership (e.g., stock certificate), lengthy payment process, and any manual intervention steps that may be necessary (e.g., review).

Financial market participants have been investing in technologies to shorten the transaction settlement time. The term *straight-through processing* (STP) is often used to designate the ideal transaction processing system that does not involve manual intervention, leading to decreased processing time, reduced operational risks, and lowered costs. Additionally, financial markets have adopted common conventions for settlement periods. For example, the current convention in securities transactions is *T+2 (trade date plus two days)*, which indicates that a security transaction should be settled two business days after the transaction initiation date. Recently, the term *same-day affirmation* (SDA) or *T0* has received attention from markets and regulators as it promotes the idea of completing the transaction verification process on the same day the transaction took place. Reducing the time of financial transaction lifecycles via STP and SDA can be quite challenging, as it requires significant investments by multiple market participants in a data infrastructure that provides real-time transaction data, while at the same time meeting the

requirements of security and reliability (luckily, this is the topic of this book!).

Settlement method

Financial market transactions can be settled differently based on the transaction type. In securities markets, the most common settlement method is *Delivery versus Payment* (DVP), which guarantees that the cash payment for securities is made either before or at the same time as the delivery of the securities. In a *Delivery versus Free* (DVF) mechanism, the delivery of securities occurs for free, for example, when delivering the collateral to a securities loan.

In payments, multiple settlement mechanisms exist. A common example is *Payment versus Payment* (PvP), widely used in Forex transactions. In a PvP system, the payment of one currency occurs only if the payment of the corresponding currency takes place. Another prominent mechanism is *Real-Time Gross Settlement* (RTGS), where funds are transferred in real time on a gross basis, meaning transactions are processed continuously throughout the day and settled individually. Conversely, in a *Deferred Net Settlement* (DNS) mechanism, transactions are accumulated over a period (e.g., end of the day) and then settled in batches at specific intervals. In DNS, only the net difference between debits and credits is settled. For example, if Citibank is to pay JPMorgan Chase \$1.5 million, and JPMorgan Chase is to pay Citibank \$1 million, a DNS system would aggregate this to a single payment of \$500,000 from Citibank to JPMorgan Chase. Conversely, an RTGS system would require two separate payments for the full amounts (\$1.5 million to JPMorgan Chase and \$1 million to Citibank).

Transaction parties

Financial market transactions are often conducted by two parties, the buyer and the seller. However, transactions are not risk free. The term *settlement risk* refers to the risk that one side of a transaction does not honor their contractual obligation, e.g., failure to pay the amount due, failure to pay on time, or failure to transfer the asset ownership. To miti-

gate such risk, a third party is often involved in financial transactions to act as a guarantor. Examples include the following:

Clearing house

Clearing houses act as intermediaries between sellers and buyers. They facilitate and guarantee a successful settlement of transactions by acting as buyers for the seller and sellers for the buyer. Examples of clearing houses include exchange clearing house divisions such as Nasdaq Clearing and CME Clearing, as well as credit card clearing houses such as Visa and Mastercard.

Central Securities Depository (CSD)

A CSD is a specialized financial institution that holds financial securities such as shares and derivatives on behalf of other institutions. Similar to a clearing house, CSDs act as a third party to financial transactions to guarantee successful settlement and transfer of security ownership against the payment of money. A prominent example of a CSD is Clearstream.

Custodian

A custodian or custodian bank is a financial institution that offers securities and post-trade services to institutional investors, asset managers, banks, and other financial institutions. Custodians provide a range of services, such as holding and safekeeping their client's securities, conducting transactions on their client's behalf, and providing clearing and settlement services.

Payment systems

Payment systems facilitate the secure transfer of funds between financial institutions while effectively managing settlement risks. The most prominent example is RTGS systems such as the Federal Reserve's Fedwire in the United States, the Bank of England's CHAPS in the UK, and the European Central Bank's T2 (part of TARGET2 services). RTGSs are typically preferred for fast transactions and reduced settlement risk. Other services allow for deferred and netted payments, such as the US Clearing House

Interbank Payments System (CHIPS). These systems are typically used for less-time-critical payments, which makes them less expensive than RTGSs.

Analytics Data

A valuable type of financial data is analytics data, which is often derived from other types of data such as fundamental, market, and transaction data. This data is typically computed using simple formulas, statistical models, machine learning techniques, and financial theories. Examples include news and market sentiment analytics (novelty, score, relevance, impact), financial risk measures (e.g., Value at Risk, option Greeks, bond duration, implied volatility), market indexes (e.g., MSCI Global Indexes), ESG (environmental, social, and governance) scores, stock analysis, company valuation, estimates (e.g., Institutional Brokers' Estimate System estimates), and competition analysis.

The main advantage of analytics data is that it offers pre-calculated patterns and signals, ready for immediate use in decision-making.

Nevertheless, the calculation methodology used by the analytics data provider might be proprietary, which makes it a mystery box.

Furthermore, as the same data becomes accessible to all market participants, maintaining a sustainable competitive advantage becomes increasingly difficult (sooner or later, someone will get the same data and replicate your strategy).

Alternative Data

Alternative data refers to a variety of nonconventional data that can be used for financial analysis, investment, or forecasting. It is regarded as nonconventional since it does not primarily originate within the traditional context of financial markets, such as trading, prices, transactions, and corporate finance operations. Examples include news, social media posts, satellite images, patents, Google trends, consumer behavior, technology analysis, political analysis, environmental and reliability scores, and many more. A large number of alternative datasets are available

through data vendors, but it is also possible to extract similar data from the internet and other public repositories.

The main advantage of alternative data lies in its novelty and diversity. A financial institution equipped with the expertise and resources to extract and clean alternative data sources stands to gain a competitive informational advantage, which can significantly improve portfolio analysis and returns. For example, one [study](#) used satellite image data to show that the number of cars in the parking lots of a sample of public US companies may be used as a predictor of financial performance.

A variety of challenges might be encountered when working with alternative data. The main issue is that these datasets are often available in an unstructured and raw format, necessitating substantial investment to structure the data and extract valuable insights. Another challenge is the lack of standardized entity identifiers and references, unlike conventional financial datasets. Moreover, alternative data can be imbalanced or incomplete because its observations and events are not consistently captured, unlike the systematic data collection for traditional datasets such as publicly traded stocks. Finally, alternative datasets can easily be biased yet provide no additional information to detect such biases.¹³

Reference Data

Financial reference data is a type of metadata used to identify, classify, and describe financial instruments, products, and entities. It is crucial for various financial operations, including transactions, trading, clearing and settlement, regulatory reporting and compliance, data consolidation, and investment.

To understand what reference data is, it's helpful first to review the concept of a financial instrument. Simply put, a financial instrument is a contract between two parties that has monetary value and can be traded on financial markets. Examples include debt instruments such as bonds and loans, equity instruments such as stocks, derivative instruments such as options and futures, and foreign exchange instruments such as currency pairs and currency futures. Importantly, each financial instrument comes

with its own set of terms and conditions that define its contractual specifications. In this context, reference data is typically used as a metadata resource to describe these terms and conditions. [Table 2-9](#) presents a few examples of reference data for several financial assets.

Table 2-9. Reference data for different asset classes

Asset category	Reference data fields
Fixed income (bonds)	• <i>Issuer information</i>: name, country, sector • <i>Security identifiers</i>: ISIN; CUSIP, etc. • <i>Instrument information</i>: issue date, maturity date, coupon rate and frequency, currency, current price, yield to maturity, accrued interest • <i>Bond features</i>: callable, putable, convertible, payment schedule, settlement terms • <i>Credit information</i>: credit rating, credit spread
Stocks	• <i>Basic information</i>: identifiers such as ticker or ISIN, company name, exchange, sector, industry • <i>Price and volume information</i>: current, open, close, high, low prices, volume, average volume • <i>Dividend information</i>: dividend yield, dividend per share, dividend payment date, ex-dividend date • <i>Corporate actions</i>: stock splits, dividend distribution, buybacks, and mergers and acquisitions • <i>Fundamental information</i>: earnings per share, price-to-earnings ratio, book value per share
Funds	• <i>Fund structure</i>: umbrella fund, subfunds, share classes • <i>Identification information</i>: fund/subfund/share class name, type, identifiers such as ISIN, inception date, currency, domiciliation • <i>Fund management</i>: fund manager, management company, custodian, distributor, investment strategy, and fees such as management and advisory fees • <i>Performance</i>: net asset value (NAV), NAV calculation frequency, historical returns, benchmark index • <i>Holdings and allocation</i>: top holdings, sector, geographical, and asset allocation

**Asset
category**

Reference data fields

- *Risk profile*: volatility, Sharpe ratio, beta, alpha, hedging policy
- *Investment, distribution, and dividends*: distribution frequency, distribution restrictions, dividend yield, subscription and redemption terms, cutoff times, minimum investment

Derivatives

- *Basic information*: instrument type, underlying asset, identifiers, classification
- *Contract specifications*: contract size, expiration date, settlement date, exercise style, strike price
- *Pricing information*: premium, mark-to-market price, bid/ask price
- *Underlying asset information*: underlying asset price, volatility, dividend yield
- *Risk metrics*: delta, gamma, theta, implied volatility, etc.
- *Trading information*: open interest, volume
- *Counterparty information*: name, credit rating, collateral requirements
- *Corporate action adjustments*: stock splits on the underlying asset

Managing reference data is one of the most outstanding challenges in financial markets. In some cases, reference data is simple. For example, a stock is a standard financial instrument widely understood to represent an ownership stake in a firm. Stock reference data includes identifiers such as its ticker, the dividend rate if it pays dividends, and the legal issuing entity. Nevertheless, exotic and more complex financial instruments have been introduced with the evolution of financial markets. This is particularly the case with financial derivative instruments. For example, identifying an option contract requires information about the underlying asset, strike price, expiration date, option type (e.g., call or put), style (American versus European), and possibly other factors such as dividend

yields or implied volatility. Furthermore, derivative instruments come in great variety, each with unique characteristics and terms. Such characteristics might even change with time, either through contractual adjustments or market events, or be customized to meet clients' specific needs.¹⁴

Many financial transactions and settlements fail due to operational errors, often linked to poor reference data. These errors can stem from inaccuracies in settlement instructions, trade-specific details, client and counterparty information, instrument data, or corporate actions data. For example, incorrect client details, mismatched account information, or erroneous security identifiers can lead to misrouted payments and failed trades. Consequently, maintaining accurate and up-to-date reference data is crucial for smooth transaction processing and minimizing operational risks.¹⁵

This has led to the development of proprietary or firm-specific reference data descriptions. As a result, market participants and regulators have struggled to agree on standard terms, definitions, and formats for financial instruments, which is the primary challenge in reference data management.

Another significant challenge with reference data is the absence of a unified identification system for financial instruments. Various financial identifiers have been created, each serving different purposes and possessing unique features. Market participants' use of different identifiers makes it increasingly difficult to match financial instruments across different identification systems. This issue is explored in greater detail in [Chapter 3](#).

Several initiatives have been launched to address the challenges of reference data in financial markets. The International Organization for Standardization [established a dedicated committee](#), ISO/TC 68/SC 8, whose scope is “standardization in the field of reference data for financial services.” In the United States, the Dodd-Frank Wall Street Reform and Consumer Protection Act of 2010 mandated that the *Office of Financial Research* (OFR) [prepare and publish a financial instrument reference](#)

[database](#). In the European Union, the *Financial Instruments Reference Data System* (FIRDS) was established to [collect and publish reference data on financial instruments](#). This system operates under Article 27 of Regulation (EU) No 600/2014 (MiFIR) and Article 4 of Regulation (EU) No 596/2014 (MAR). Managed by the European Securities and Markets Authority (ESMA), FIRDS ensures the availability and transparency of financial instrument data, aiding in regulatory compliance and market supervision.

In the financial data market, several commercial reference data solutions stand out. Notable examples include Bloomberg's [reference data](#) and LSEG's [reference data](#) offerings. Reference data could be offered in a form that matches the requirements set by specific regulatory regimes, such as MiFID II, Europe's Securities Financing Transactions Regulation (SFTR), and the Basel Committee's Fundamental Review of the Trading Book (FRTB). Additionally, some reference data services focus on financial entities, such as [SwiftRef](#), which provides comprehensive international payment reference data. SwiftRef offers detailed information on BICs (Business Identifier Codes), IBAN (International Bank Account Number) validation, and various other identifiers crucial for identifying entities involved in global payments.

In [Chapter 3](#), we will explore the most essential types of reference data, focusing specifically on financial security identifiers.

Entity Data

Entity data includes information about corporate entities and their characteristics. It is frequently used alongside reference data to offer more detailed insights into a specific entity. Examples of entity data elements include the company name, identifiers, year of establishment, legal corporate form, ownership structure, sector classification, associated individuals, credit rating, ESG score, risk exposures, major corporate events, and more.

A number of commercial entity datasets are available, such as LSEG's [legal entity data and services](#), Moody's [Orbis database](#), and SwiftRef's [entity](#)

[data](#). These datasets are used for many useful purposes such as corporate finance analysis, risk management (e.g., supplier or credit risk), and compliance and financial crime prevention through anti-money laundering (AML), know your customer (KYC), and client due diligence (CDD).

Benchmark Financial Datasets

As you have seen from our discussion thus far, financial data is abundantly available from multiple sources and in a variety of types and structures. To extract analytical and economic value from this data, commercial vendors, financial institutions, and researchers create *financial datasets*. I define a financial dataset as a bundled collection of variables and data points that provide information on a specific financial entity or topic, such as loans, stock prices, index prices, bond markets, and derivative markets. A financial dataset can include a mix of data types, such as fundamentals, market data, transactions, analytics, reference data, and entity data.

A considerable number of financial datasets already exist in the market, and new datasets are continuously being created and published. Next, I'll provide an overview of some of the world's most trusted and used financial datasets.

Center for Research in Security Prices

Financial datasets provided by the Center for Research in Security Prices (CRSP) are among the most prominent and trusted sources of market data. In particular, the [CRSP US Stock dataset](#) provides comprehensive daily and monthly stock market data on over 32,000 securities listed on US stock exchanges such as the NYSE and NASDAQ and on a broad range of market indexes. The CRSP US Stock dataset contains information on price and quote data as well as external identifiers, shares outstanding, market capitalization, delisting information, and corporate action details.

Compustat Financials

[Compustat Financials](#), also called Compustat Fundamentals or Compustat Global, is the world benchmark dataset for company fundamentals. It provides standardized information on more than 80,000 international public companies. Compustat provides point-in-time snapshots of fundamental data, allowing researchers to conduct reliable historical and back-testing analyses. The dataset is quite comprehensive, with over 3,000 fields conveying information on financial statements, ratios, corporate actions, industry specifications, identifiers, and more.

Trade and Quote Database

The daily [Trade and Quote \(TAQ\) dataset](#) provides daily high-frequency data on all trades and quotes that take place on the NYSE, NASDAQ, and other regional exchanges. TAQ data is available starting from 2003. The distinguishing feature of TAQ is the time precision it offers: seconds (HHMMSS) since 1993, milliseconds (HHMMSSxxx) since 2003, microseconds (HHMMSSxxxxxxx) since 2015, and nanoseconds (HHMMSSxxxxxxxxxxx) since 2016.

Institutional Brokers' Estimate System

The [Institutional Brokers' Estimate System \(I/B/E/S\)](#) is a database maintained by LSEG, serving as the market benchmark for stock analysts' earnings estimates for publicly traded companies. Over 950 firms and more than 19,000 analysts from 90+ countries regularly contribute to the I/B/E/S. The database provides extensive coverage of over 60,000 companies, with data available from 1976 for North America and from 1987 for international markets.

IvyDB OptionMetrics

[IvyDB](#) is the market benchmark dataset for historical equity and index options data. The most popular version of IvyDB is the US database, but since 2008, IvyDB Europe has been available as well. IvyDB provides a

rich set of options data fields, such as daily quotes, identifiers, volume, computed implied volatility, option Greeks, open interest, interest rates, maturity, exercise, and exercise price. Information on the underlying instruments is also available, including closing prices, dividends, and corporate action information.

Trade Reporting and Compliance Engine

[Trade Reporting and Compliance Engine \(TRACE\)](#) is a financial dataset on bond and fixed-income transactions. TRACE represents an indispensable data source as most bond transactions happen OTC, making it a less transparent market than centrally exchanged instruments such as stocks.

TRACE itself is a program created by the Financial Industry Regulatory Authority (FINRA) to enable market participants who are FINRA members to report their fixed-income OTC transactions. FINRA members are required to report their transactions within 15 minutes of execution, which is then made available in real time to TRACE subscribers. Data available via TRACE includes fields such as execution time, price, volume, and bond yield.

Orbis Global Database

[Orbis](#) is the industry's primary resource for global entity data. Published by Moody's Analytics, it contains information on more than 450 million private and listed companies worldwide, offering detailed financial information for many of them.

In addition to its global coverage, Orbis provides comparable information on company ownership structure and financial strength metrics. Orbis collects data from more than 170 data providers and hundreds of its own data sources. The data is further enriched and standardized to enable easy querying, analysis, and comparison.

SDC Platinum

[SDC Platinum](#), offered by the London Stock Exchange Group (LSEG), is a premier source of comprehensive global corporate finance and market deal event data. It provides detailed information on various financial transactions, including mergers and acquisitions, alliances, private equity, venture capital, new issues, leveraged buyouts, and syndicated loans, among many others.

Standard & Poor's Dow Jones Indices

[Standard & Poor's Dow Jones Indices \(SPDJI\)](#) is a leading global index provider and a primary source of historical index data, offering a wide range of indexes across various markets, including equities, derivatives, fixed income, and commodities. Examples include the S&P 500, S&P MidCap 400, and S&P SmallCap 600, which are widely recognized as leading indicators of US equity market performance. S&P DJI provides detailed data features such as index names, constituents and their weights, closing prices, market capitalization, constituent company information, and index-related events.

Alternative Datasets

The datasets illustrated so far are general in scope and generated through the traditional mechanisms of financial markets. Interestingly, alternative, more specialized datasets have been gaining popularity among market participants. Let's look at a few examples.

BitSight Security Ratings

[BitSight Security](#) is a world leader in cybersecurity rating and related analytics. BitSight ratings convey comparable insights and visibility into a company's cybersecurity risk. It provides adaptive ratings correlated with the changing ransomware risk landscape. BitSight ratings are calculated using objective and observable factors such as server software, open ports, TLS/SSL (Transport Layer Security/Secure Sockets Layer) certificates and configuration, web application headers, and system security.

Global New Vehicle Registrations

The [Global New Vehicle Registrations dataset](#), offered by S&P Global Mobility, provides daily information and analysis on vehicle registrations from more than 150 countries, 350 brands, multiple fuel types (diesel, petrol, etc.), and body types (e.g., car, van, SUV). The dataset provides valuable information that can be used to analyze trends in the automotive market, such as the transition to electric vehicles.

Weather Source

The [Weather Source dataset](#) provides hourly and daily weather-related data for a large number of locations worldwide. Weather Source collects and standardizes weather data from many input sources and provides weather insights relevant to different businesses.

Patent data

Patent data is unstructured data that conveys information on patent details such as the inventor and assignee name, related patent citations, patent abstract, patent summary, detailed description, and claims. This data is often used to understand technological innovation problems. In recent years, the use of patent data has gained increasing importance in financial analysis. One of the primary sources of patent data is the [United States Patent and Trademark Office \(USPTO\)](#), which provides public access to detailed patent and trademark information. Another useful source is Google Patents, which aggregates patent data from a variety of sources, including the USPTO, and makes it easily searchable via its search engine.

In conclusion, it's important to keep in mind that the number of datasets created and used by financial markets is immense, and we have only scratched the surface in this section. As a financial data engineer, one of the most valuable skills that you can develop is the ability to search and navigate the financial data landscape and identify the right dataset for a particular business problem.

Summary

This chapter provided an overview of the financial data landscape, which we can summarize as follows:

- Classifying and explaining the sources of financial data
- Distinguishing between the different structures used to represent financial data
- Illustrating the main types of financial data generated by the various market activities
- Providing a short list of benchmark datasets widely recognized among market participants and researchers

Now that you have an understanding of what financial data engineering entails and the intricacies of the financial data ecosystem, we'll shift gears with the next few chapters, where you will learn about specific financial data engineering topics. The following three chapters will address in depth the following problems, which have been selected based on their prominent importance for financial markets:

- Financial identification systems and their main features and challenges ([Chapter 3](#))
- The process and methods for financial entity recognition and resolution ([Chapter 4](#))
- Financial data governance through data quality, integrity, privacy, and security ([Chapter 5](#))

Let's keep going!

¹ If interested in this topic, I recommend Itay Goldstein and Liyan Yang's article, ["Information Disclosure in Financial Markets"](#), *Annual Review of Financial Economics* 9 (November 2017): 101–125, and Anat R. Admati and Paul Pfleiderer's ["Forcing Firms to Talk: Financial Disclosure Regulation and Externalities"](#), *The Review of Financial Studies* 13, no. 3 (July 2000): 479–519.

- 2** If interested in learning more about portfolio theory, I recommend the excellent book by Jack Clark Francis and Dongcheol Kim, *Modern Portfolio Theory: Foundations, Analysis, and New Developments* (Wiley, 2013).
- 3** For a good introduction to network science, I highly recommend Mark Newman's book *Networks*, 2nd ed. (Oxford University Press, 2018).
- 4** For more on this topic, see Celina M.H. de Figueiredo's article "[The P versus NP—Complete Dichotomy of Some Challenging Problems in Graph Theory](#)", *Discrete Applied Mathematics* 160, no. 18 (December 2012): 2681–2693.
- 5** For a good introduction to the different sources of financial text data, I recommend Mirjana Pejić Bach, Živko Krstić, Sanja Seljan, and Lejla Turulja's article, "[Text Mining for Big Data Analysis in Financial Sector: A Literature Review](#)", *Sustainability* 11, no. 5 (January 2019): 1277.
- 6** More on this topic in Marjan Van de Kauter, Diane Breesch, and Véronique Hoste's "[Fine-Grained Analysis of Explicit and Implicit Sentiment in Financial News Articles](#)", *Expert Systems with Applications* 42, no. 11 (July 2015): 4999–5010.
- 7** To see a real example, check out the [information provided by Yahoo Finance for Google stock](#).
- 8** For a more comprehensive list, check CME Group's [list of match algorithms](#).
- 9** For a deeper explanation of the order book mechanism, I recommend Jean-Philippe Bouchaud, Julius Bonart, Jonathan Donier, and Martin Gould's book, *Trades, Quotes and Prices: Financial Markets Under the Microscope* (Cambridge University Press, 2018).
- 10** For an excellent reference on this topic, I recommend Aswath Damodaran's *Investment Valuation: Tools and Techniques for Determining the Value of Any Asset*, 3rd ed. (Wiley, 2012).
- 11** To calculate the NBBO, stock exchanges report their best bid/ask prices to a system called the Securities Information Processor (SIP), which aggregates all the quotes in a single NBBO and releases it to the market.
- 12** Readers interested in a comprehensive coverage of the financial transaction life-cycle can read the excellent book by Robert P. Baker, *The Trade Lifecycle: Behind the Scenes of the Trading Process* (Wiley, 2015).

- 13** For more on this topic, see Ashby Monk, Marcel Prins, and Dane Rook's article, ["Rethinking Alternative Data in Institutional Investment"](#), *Journal of Financial Data Science* 1, no. 1 (Winter 2019): 14–31.
- 14** For example, consider a stock option with a strike price of \$100 and 100 shares underlying it. If a two-for-one stock split event takes place, managing reference data involves adjusting the option specifications to reflect 200 shares with a new strike price of \$50, ensuring all related systems and records are updated accurately.
- 15** A more detailed discussion of this problem can be found in Allan D. Grody, Fotios Harmantzis, and Gregory J. Kaple's article, ["Operational Risk and Reference Data: Exploring Costs, Capital Requirements and Risk Mitigation"](#), *Journal of Operational Risk* 1, no. 3 (2006).